Brain Tumor MRI Segmentation Method Based on Improved Res-UNet

Xue Li, *Student Member, IEEE*, Zhenqi Fang, Ruhua Zhao, *Graduate Student Member, IEEE*, and Hong Mo, *Member, IEEE*

***Abstract*—Automatic segmentation of MRI images is crucial for diagnosis and evaluation of brain tumors. However, significant variability in brain tumor shape, uneven spatial distribution, and intricate boundaries bring challenges, which lead information loss and decreased accuracy during segmentation. To solve these problems, an improved Res-UNet network employing attention-guided and scale-aware strategies is proposed. First, a module that employs attention mechanisms and features fusion is incorporated to catch relatively important contextual information. Secondly, a module designed to retrieve hidden contextual information and dynamically aggregate multi-scale features is integrated into the bottom layer of the network, which facilitates feature acquisition and enhancement at multiple scales. Finally, the results show that the method achieves a Dice similarity coefficient of 92.24% in whole tumor region, which is an improvement of about 4% compared to the pre-improved Res-UNet network.**

***Index Terms*—Brain tumor segmentation, Res-UNet, feature fusion, feature enhancement.**

I. Introduction

GLIOMA is the most common malignant primary brain tumor. Statistically, the average overall survival of patients with gliomas is 12 to 15 months [1]. Magnetic resonance imaging (MRI) is an imaging method that is free of ionising radiation to the body and is capable of mapping the internal structure of the human brain [2]. In clinical surgery, doctors manually segment brain tumors with MRI images, which is not only limited by the quality of MRI images, but also depends on the clinical experience of doctors. Thus, it is essential to design a fully automated method for brain tumor segmentation.

However, this is still a challenging task. Firstly, due to the wide distribution of glial cells, the location of segmentation is uncertain. Second, the tumour morphology of the same patient varies greatly at different time points, and the contrast of the MRI images was low, which increased the difficulty of segmenting tissue boundaries. Finally, manually annotated data relies heavily on the physician's experience and is prone to annotation bias. To address these issues, brain tumour segmentation methods have been improved. These include traditional methods and methods based on deep learning. The traditional segmentation methods are the following: The thresholding algorithm is the typical method used to segment brain tumour MRI images [3]. The edge detection algorithm determines the tumour edge pixels by the detection of the intensity change in the image [4]. The region growing algorithm integrates neighbouring pixels with similar features into a single region [5]. The watershed segmentation method combined with the expectation maximisation algorithm was able to segment grey matter and cerebrospinal fluid in MRI images of the brain [6].

However, these methods are simple in structure and cannot accomplish accurate segmentation alone. Deep learning based segmentation methods have made significant advancements in the field of image segmentation. Common segmentation algorithms are convolutional neural network (CNN) based algorithms and recurrent neural network (RNN) based algorithms [7], [8]. Developed from CNN, full convolution network (FCN) is the first end-to-end full convolution network for the prediction of individual pixels, which realizes arbitrary size input [9]. The U-Net network is improved from the FCN, which make full use of contextual features [10]. The Seg-Net model is a derivative of the FCN architecture, which solves the problem that the skip connection in FCN only replicates the encoder features and the segmentation is rough [11]. Residual blocks were used instead of original convolution blocks in U-Net to solve the degradation problem, resulting in the Res-UNet [12]. Hash-based DenseHashNet image retrieval algorithm for faster and more accurate retrieval of relevant medical images [13]. NCS-FPN is a non-local channel and spatial attention feature pyramid network with enhanced multi-scale feature extraction [14]. DRINet addresses the problem that standard convolutional layers do not perform well when dealing with different classes with similar features [15]. The deep and fuzzy-based learning hierarchical fusion model addresses mammography image pixel segmentation and disease classification [16]. DeepLab V3 combined with Google Street View (GSV) imagery enables a more efficient and safe curb environment [17].

Although the segmentation performance of these networks has been improved, the efficiency of extracting multi-scale contextual information still needs to be improved. The challenges include the impact of noise, motion artefacts, and magnetic field aberrations on MRI images of brain tumours [18], which augment the complexity of segmentation.

Manuscript received 14 December 2023; accepted 27 December 2023. Date of publication 2 January 2024; date of current version 25 July 2024. *(Corresponding author: Hong Mo.)*

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Digital Object Identifier 10.1109/JRFID.2023.3349193

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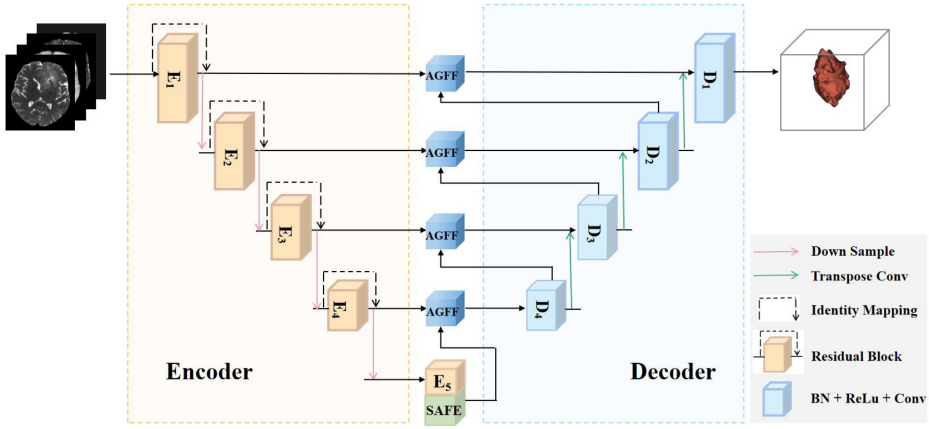

Fig. 1. Network architecture.

Furthermore, brain tumours differ greatly in terms of type, location, and shape [19]. Additionally, the training of deep learning models is limited due to the scarcity of brain tumour MRI datasets.

Thus, we proposed an improved 3D network for brain tumor segmentation based on Res-UNet architecture. Portions of this work were presented at the IEEE International Conference on Digital Twins and Parallel Intelligence in 2023 [20]. First, a module that employs attention mechanisms and feature fusion (AGFF) is used to replace the jump connection layer of U-Net network, and the most valuable contextual information is selected for fusion. Secondly, a module designed to enhance multiple scale features (SAFE) is added to the lowest layer of the network to retrieve hidden contextual data, which improve Network architecture the network's ability to represent features. Finally, the results show that the method outperforms the pre-improvement algorithm in brain tumor segmentation, enhancing the ability to effectively fuse contextual information.

II. MRI Segmentation Algorithm of Brain Tumor Based on Improved Res-UNet

A. Network Structure

To increase the efficiency and accuracy of brain tumour MRI image segmentation, we have merged the U-Net network and ResNet for feature fusion and multi-scale feature adaptive selection at the jump connection layer and the bottom of the network, as depicted in Fig. 1.

Deepening the network structure enhances the semantic understanding of the model. However, the training of networks that are excessively deep poses problems like gradient vanishing and gradient explosion. These issues can be overcome with ResNet, which is used to train a highly deep network. The 3D Res-UNet network employs similar concepts as ResNet by substituting the encoder section of the U-Net network with a residual block. Fig. 2 displays the residual block module within ResNet. This module comprises convolutional layers, BN, ReLU that overlays the output from the image input via the two branches. This operation enables the data to flow through the network across layers without introducing extra parameters. Technical abbreviation explanations have been provided where necessary. Furthermore, the instability of the network's training is attributed to varied input distributions across layers. To remedy this, the BN layer normalises input to each layer, thereby improving convergence speed and the overall stability of the training process.

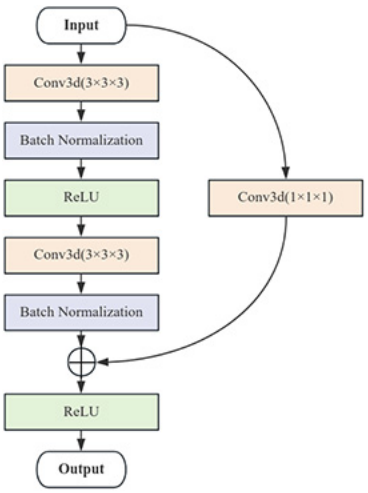

Fig. 2. Residual block structure.

B. Attention-Guided Feature Fusion Module

Low-level features are derived from shallow networks and possess plentiful spatial information, allowing for precise identification of high-level features, but lacking semantic information. Conversely, high-level features are obtained from the deep network and possess ample semantic information, aiding in identification of semantically relevant location of low-level features, while lack spatial information. Reducing low-level noise and integrating valuable contextual information are key factors in improving the accuracy of complex boundary segmentation for brain tumors. Therefore, a feature fusion module is introduced which effectively extracts information from neighbouring layers.

As shown in Fig. 3, the advanced feature $F_h^{(i+1)}$ is upsampled to the same size as the low-level feature $F_l^{(i)}$ (where

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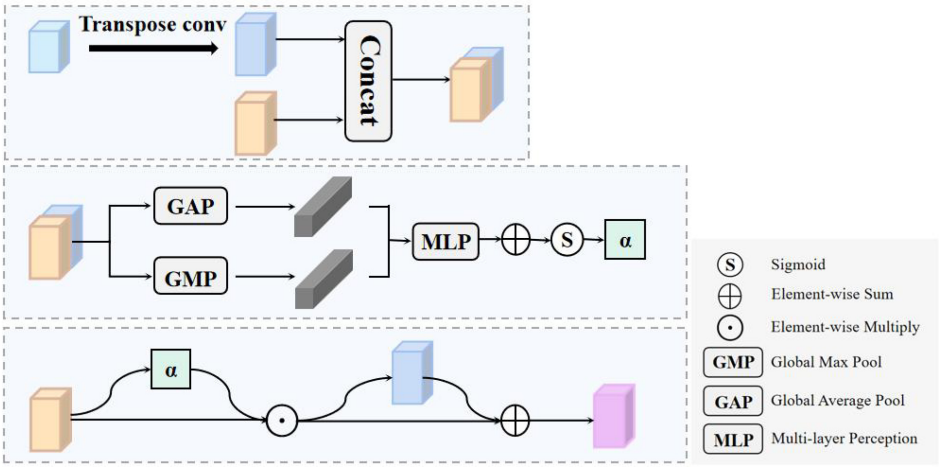

Fig. 3. AGFF structure.

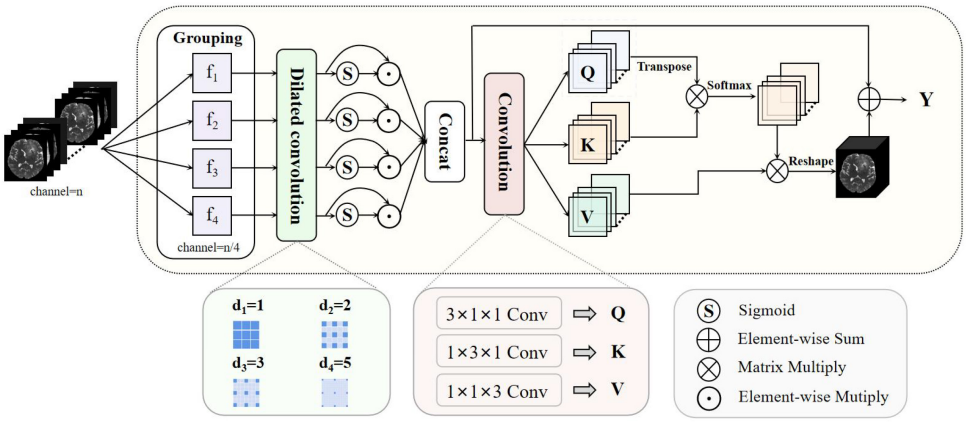

Fig. 4. SAFE structure.

i = 1, 2, 3, 4), after which the two features are concatenated along the channel dimension, as in:

$$F_c^{(i)} = C\left(F_l^{(i)}, f_u\left(F_h^{(i+1)}\right)\right) \tag{1}$$

where f_u represents the up-sampling operation and C stands for the channel splicing operation.

Secondly, the SE-block performs global average pooling of features to compress the global spatial information into channel descriptions. The compressed vectors learn the weights of each channel through the fully connected layer to motivate the useful feature channels and suppress the useless ones. After two pooling operations respectively, the shared multi-layer perceptron (MLP) and sigmoid activation function nonlinearly transforms and combines the feature vectors to obtain the channel attention coefficients, defined as:

$$\alpha = f_\delta\left\{f_{mlp}\left(f_{gap}\left(F_c^{(i)}\right)\right) + f_{mlp}\left(f_{gmp}\left(F_c^{(i)}\right)\right)\right\} \tag{2}$$

where f_{gmp} stands for the global pooling with maximum operation, f_{gap} stands for the global pooling with average operation, f_{mlp} stands for the shared multi-layer perceptron, and f_δ represents the activation function sigmoid.

Since the lower-level features contain more details, multiplication of the channel attention coefficient α and the low-level features to obtain the re-weighted vector. Subsequently, the re-weighted vector is summed with the high-level features using the ResNet principle, as in:

$$F^{(i)} = \alpha \odot F_l^{(i)} + F_h^{(i)} \tag{3}$$

where $\odot$ stands for element multiplication.

C. Scale-Aware Feature Enhancement Module

In the segmentation process, enhancing the network feature representation ability can optimize the generalization ability and performance of the model, so that the model can better distinguish between normal and diseased tissues and improve the accuracy and stability of segmentation. The SAFE module is embedded in the underlying network. Its functions include feature extraction across multiple scales and adaptive feature aggregation. As shown in Fig. 4, the module first divides the feature graph generated by Encoder5 into four groups on average according to the number of channels, while keeping the feature map size unchanged, obtaining f_i, $i \in \{1, 2, 3, 4\}$. Then, we use dilated convolutions with different dilation factors to generate feature maps with varying receptive fields,

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making the network fully aware of features across multiple scales, as in:

$$f_i' = f_\delta\left(f_c^{d_i}(f_i)\right) \odot f_c^{d_i}(f_i) \tag{4}$$

where $f_c^{d_i}$ stands for the dilation convolution with the dilation factor d_i, $i = (1, 2, 3, 5)$, and f_δ stands for the activation function sigmoid. Subsequently, we concatenate four sets of feature maps with different receptive fields and perform adaptive feature aggregation, as in:

$$F = C\left(f_1', f_2', f_3', f_4'\right) \tag{5}$$

where C represents the channel splicing operation.

In medical imaging, cutting the human body into sagittal, coronal and transverse planes facilitates the observation and analysis of structures and organs inside the human body. Based on this, we apply convolution operations on feature F with kernel sizes of $3 \times 1 \times 1$, $1 \times 3 \times 1$, $1 \times 1 \times 3$, respectively. Then, we pass the feature maps through BN layer and ReLU activation to obtain Q, K, and V. In the feature F, the dimension C corresponds to the channel, H represents the height, W denotes the width, and D signifies the depth. Subsequently, softmax is applied to the matrix obtained by matrix multiplication of Q^T and K. In this process, Q^T and K are medical image data from the sagittal and coronal planes of the human body. Matrix multiplication allows us to combine and encode these feature matrices to produce a new matrix that contains the relationships between the features. The softmax operation converts the value of each element into a probability value, which enables the model to prioritize significant features. Finally, it is matrix multiplied with V to obtain the features enhanced based on voxel attention mechanism. Then its shape is reshaped as $R^{C\times H\times W\times D}$ and multi-scale features are aggregated by combining the idea of residual network. The process is defined as:

$$Y = F + softmax\left(Q^T \otimes K\right) \otimes V \tag{6}$$

where $\otimes$ stands for matrix multiplication.

D. Loss Function

The loss function is used as a constraint to optimize the network training parameters, and it measures the gap between the network prediction and the true label, as follows:

$$L = (1 - \beta)L_{ce} + \beta L_{dice} \tag{7}$$

where L_{ce} is the cross-entropy loss function, L_{dice} stands for Dice loss function, and the parameter β is used to adjust the weight balance between the two functions. The equations for both functions can be expressed as follows:

$$L_{dice} = -\frac{1}{K}\sum\nolimits_{k\in K}\frac{2\sum\nolimits_{i\in I}u_i^k v_i^k}{\sum\nolimits_{i\in I}u_i^k + \sum\nolimits_{i\in I}v_i^k} \tag{8}$$

$$L_{ce} = -\frac{1}{K}\sum_{k=1}^{K}\left[u_i^k \log v_i^k + \left(1 - u_i^k\right)log\left(1 - v_i^k\right)\right] \tag{9}$$

where u_i^k and v_i^k are the network predictions and the true label value containing only 0 and 1 respectively, $i \in I$ is the number of pixels in the training batch, and $k \in K$ is the sample category.

The utilization of both loss functions can reduce the problems caused by the imbalance of segmentation classes and accelerate the learning speed of the network. After testing, when β is 0.6, the network loss is minimized.

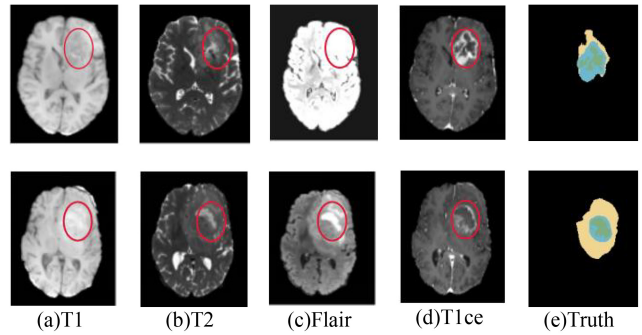

Fig. 5. MRI images of four modalities and real label images of the patient's brain tumor.

III. Experiment

A. Dataset

The dataset used in this paper is from the brain tumor MRI image public data set BraTS2021 provided by MICCAI. Each case in this data set contains brain MRI images of four modalities and true label marked by doctors. Each modality and true label consists of 155 scanned images with varying layers, each of size 240×240. The main segmentation task of the experiment in this paper is to accurately segment three brain tumor tissue regions in the patient's MRI images, which are: whole tumor regions (WT), which contain all types of tumor tissues; tumor core region (TC), which contains gangrenous tissue and enhanced tumor tissue; enhance tumor region (ET), which mainly enhances the tumor tissue. Fig. 5 illustrates the four modalities of brain tumour MRI images of two patients and the expertly annotated true label segmented images.

For the purpose of enhancing image contrast and enlarging tumor ratio, before the network training, we merge the images of the four modalities into a 4D image (C×W×H×D, C=4) and save the segmentation label as a file. The gray level of each image is standardized, but the background area is preserved as 0. Then the image is cropped to obtain a brain tumor image of size 160×160×128, and the training dataset is expanded by data enhancement methods including Gaussian blur, random flip and pan.

B. Evaluation Index

Three classical evaluation metrics were used to assess the results: Dice similarity coefficient (DSC), precision and recall. Among them, DSC is used to measure the resemblance between the network prediction and the real labels annotated by experts, recall is used to measure the detection rate of tumor tissue, and precision is used to measure the accuracy of prediction. The three indices range from 0 to 1, with higher

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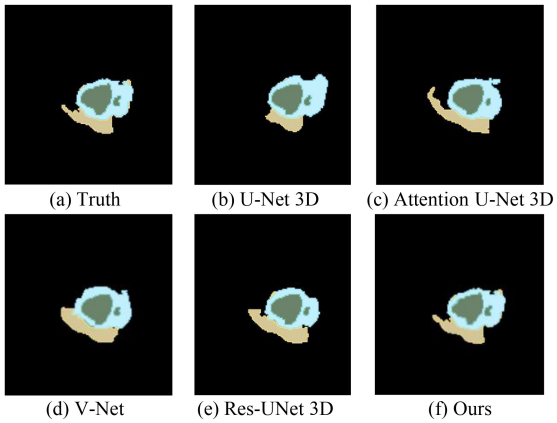

Fig. 6. Comparison of outcomes across various segmentation methods.

values indicating greater accuracy in segmenting. The formula is as follows:

$$DSC = \frac{2 \cdot |TP|}{2 \cdot |TP| + |FN| + |FP|} \tag{10}$$

$$Recall = \frac{|TP|}{|TP + FN|} \tag{11}$$

$$\Pr ecision = \frac{|TP|}{|FP + TP|} \tag{12}$$

In the above formula, we assume that brain tumor tissue is a positive sample and normal brain tissue is a negative sample. T represents the tumor tissue region annotated by experts, and P is the predicted tumor tissue region. TP represents the true positive of the pixel when it is judged to be positive and in fact positive, and FP represents the false positive of the pixel when it is actually negative. Similarly, TN refers to a sample that was correctly identified as negative when it was actually negative, and FN refers to a sample that was incorrectly identified as negative when it was actually positive.

C. Analysis of Experimental Results

To verify the superiority of our proposed method, we compared it with U-Net 3D, Attention U-Net 3D, V-Net, and Res-UNet 3D. The experiments were conducted using an NVIDIA 3090 GPU with 24GB of video memory in the PyTorch framework. Additionally, the learning rate was set to 1e-3 and the batch size to 4. The experiments used the Adam optimizer.

Fig. 6 presents a comparative analysis of segmentation outcomes for a patient. The results from the U-Net 3D segmentation show that some tumour tissue was not detected in the WT. The WT of Attention U-Net 3D is closer to the true label, but the ET and TC segmentation are not delicate enough. The segmentation shape of V-Net and Res-UNet 3D inside the tumor is close to the true label, but it still falls short on the whole. Our proposed method is very close to the true label both from the whole tumor area and the internal tumor area.

As shown in the table above, the results of the evaluation of the performance indicators DSC, recall and precision for each model in the segmentation process are presented separately. The segmentation method proposed in this paper is superior to U-Net 3D, Attention U-Net 3D, V-Net and Res-UNet 3D networks in the similarity between the predicted results and the true labels, the detection rate of tumor tissue and the accuracy of prediction.

From the table, it can be seen that Attention U-Net 3D and V-Net are better than U-Net 3D in the segmentation of the three regions, and Res-UNet 3D performs the best in the recall rate of the tumor compared with the previous two networks. The proposed method achieves 92.24% DSC, 90.42% recall and 91.57% precision on WT, which achieves the highest segmentation accuracy among the above methods.

TABLE I
DICE INDEX EVALUATION RESULTS OF FIVE MODELS

Network	DSC		
	WT	ET	TC
U-Net 3D	0.7294	0.7427	0.7231
Attention U-Net 3D	0.7921	0.7864	0.7618
V-Net	0.8258	0.8417	0.8325
Res-UNet 3D	0.8737	0.8618	0.8574
Ours	**0.9224**	**0.8921**	**0.8949**

TABLE II
RECALL INDEX EVALUATION RESULTS OF FIVE MODELS

Network	Recall		
	WT	ET	TC
U-Net 3D	0.7315	0.7254	0.7182
Attention U-Net 3D	0.7745	0.7863	0.7559
V-Net	0.8197	0.8204	0.8246
Res-UNet 3D	0.8792	0.8753	0.8769
Ours	**0.9042**	**0.9027**	**0.8965**

TABLE III
PRECISION INDEX EVALUATION RESULTS OF FIVE MODELS

Network	Precision		
	WT	ET	TC
U-Net 3D	0.7362	0.7425	0.7238
Attention U-Net 3D	0.7964	0.8031	0.7926
V-Net	0.8229	0.8348	0.8294
Res-UNet 3D	0.8767	0.8713	0.8725
Ours	**0.9157**	**0.9021**	**0.8975**

IV. CONCLUSION

In this paper, a segmentation method based on the improved Res-UNet 3D network is proposed to address the problems of information loss and low segmentation accuracy. First, the network is deepened by using a residual module while maintaining a small number of parameters. Secondly, the attention-guided feature fusion module is incorporated to efficiently select the contextual information. Finally, the feature enhancement module that can fully perceive feature cross

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multiple scale is used to extract hidden information and aggregate it to strengthen the feature representation capability of the network. The results show that the proposed method obtains higher segmentation accuracy than other classical networks.

Nevertheless, brain tumour growth and development is a complex process, and its dynamic changes cannot be fully understood by image segmentation alone. Therefore, we consider combining digital twin technology to establish a numerical model for predicting tumour growth [21]. In addition, we consider using interval type-2 fuzzy sets and fuzzy comprehensive evaluation methods to establish a risk assessment model [22], [23], [24], [25], which can help doctors develop more accurate treatment plans.

REFERENCES

[1] A. Upton and T. N. Arvanitis, "Using evolutional properties of gene networks in understanding survival prognosis of glioblastoma," *IEEE J. Biomed. Health Inform.*, vol. 18, no. 3, pp. 810–816, May 2014.

[2] J. T. Vaughan et al., "Current and future trends in magnetic resonance imaging (MRI)," in *IEEE MTT-S Int. Microw. Symp. Tech. Dig.*, San Francisco, CA, USA, 2006, pp. 211–212.

[3] N. Zhao, S. Sui, and P. Kuang, "Research on image segmentation method based on weighted threshold algorithm," in *Proc. 12th Int. Comput. Conf. Wavelet Active Media Technol. Inf. Process.*, Chengdu, China, pp. 307–310, 2015.

[4] Y. Zhao and J. Chang, "Analysis of image edge checking algorithms for the estimation of pear size," in *Proc. Int. Conf. Intell. Comput. Technol. Autom.*, 2010, pp. 663–666.

[5] W. Cui, Z. Guan, and Z. Zhang, "An improved region growing algorithm for image segmentation," in *Proc. Int. Conf. Comput. Sci. Softw. Eng.*, 2008, pp. 93–96.

[6] G. R. Kwon, D. Basukala, S. W. Lee, K. H. Lee, and M. Kang, "Brain image segmentation using a combination of expectation-maximization algorithm and watershed transform," *Int. J. Imag. Syst. Technol.*, vol. 26, no. 3, pp. 225–232, 2016.

[7] M. Savithadevi, "Convolutional neural network based retinal vessel segmentation," *Comput. Sci. Eng.*, vol. 12, no. 6, pp. 135–144, 2022.

[8] H. Kuehne, A. Richard, and J. Gall, "A hybrid RNN-HMM approach for weakly supervised temporal action segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 765–779, Apr. 2020.

[9] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 3431–3440, 2015.

[10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Interv.*, 2015, pp. 234–241.

[11] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[12] X. Xiao, S. Lian, Z. Luo, and S. Li, "Weighted res-UNet for high-quality retina vessel segmentation," in *Proc. Int. Conf. Inf. Technol. Med. Educ. (ITME)*, Hangzhou, China, 2018, pp. 327–331.

[13] C. Liu et al., "DenseHashNet: A novel deep hashing for medical image retrieval," *IEEE J. Radio Freq. Identif.*, vol. 6, pp. 697–702, Sep. 2022.

[14] X. Wu et al., "A non-local attention feature fusion network for multiscale object detection," *IEEE J. Radio Freq. Identif.*, vol. 6, pp. 733–738, Oct. 2022.

[15] L. Chen, P. Bentley, K. Mori, K. Misawa, M. Fujiwara, and D. Rueckert, "DRINet for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 37, no. 11, pp. 2453–2462, Nov. 2018.

[16] T. Shen, J. Wang, C. Gou, and F.-Y Wang, "Hierarchical fused model with deep learning and type-2 fuzzy learning for breast cancer diagnosis," *IEEE Trans. Fuzzy Syst.*, vol. 28, no. 12, pp. 3204–3218, Dec. 2020.

[17] H. Hao and Y. Wang, "Smart curb digital twin: Inventorying curb environments using computer vision and street imagery," *IEEE J. Radio Freq. Identif.*, vol. 7, pp. 168–172, 2023.

[18] M. F. Safdar, S. S. Alkobaisi, and F. T. Zahra, "A comparative analysis of data augmentation approaches for magnetic resonance imaging (MRI) scan images of brain tumor," *Acta Inf. Med.*, vol. 28, no. 1, pp. 29–36, 2020.

[19] F. Kevin et al., "Unsupervised resolution of histomorphologic heterogeneity in renal cell carcinoma using a brain tumor-educated neural network," *JCO Clin. Cancer Inform.*, vol. 4, pp. 811–821, Sep. 2020.

[20] X. Li, Z. Q. Fang, R. H. Zhao, and H. Mo, "MRI image segmentation of brain tumor based on Res-UNet," to be published.

[21] T. Azfar, J. Weidner, A. Raheem, R. Ke, and R. L. Cheu, "Efficient procedure of building university campus models for digital twin simulation," *IEEE J. Radio Freq. Identif.*, vol. 6, pp. 769–773, Oct. 2022.

[22] F.-Y Wang and H. Mo, "Some fundamental issues on type-2 fuzzy sets," *Acta Automatica Sinica*, vol. 43, no. 7, pp. 1114–1141, 2017.

[23] H. Mo, Y. Meng, F.-Y Wang, and D. Wu, "Interval type-2 fuzzy hierarchical adaptive cruise following-control for intelligent vehicles," *IEEE/CAA J. Automatica Sinica*, vol. 9, no. 9, pp. 1658–1672, Sep. 2022.

[24] H. Mo, F.-Y Wang, M. Zhou, R. Li, and Z. Xiao, "Footprint of uncertainty for type-2 fuzzy sets," in *Inf. Sci.*, vol.272, pp. 96–110, Jul. 2014.

[25] H. Mo, H. Hu, J. Hu, Y. Li, X. Wang, and F.-Y Wang, "Interval type-2 fuzzy risk evaluation and prevention for parallel breast cancer treatment system," *IEEE Trans. Comput. Soc. Syst.*, vol. 10, no. 2, pp. 673–685, Apr. 2023.

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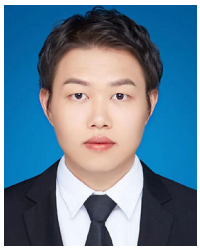

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